

STEPHENVILLE CLARK, TX, USA

WMO: 722600

Lat: 32.215N

Lon: 98.178W

Elev: 1321

StdP: 14.01

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03969

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.8	26.9	8.5	8.9	38.9	11.8	10.5	40.2	24.8	46.9	22.6	45.8	7.0	350	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.8	99.6	70.4	97.5	70.9	95.3	71.0	75.6	87.7	74.9	87.1	74.1	86.3	7.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.5	126.3	80.2	72.0	124.4	79.7	71.1	120.3	78.9	40.1	87.8	39.2	87.1	38.5	86.0	95.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	18.6	16.7	DB	16.1	102.6	4.0	2.4	13.3	104.4	11.0	105.8	8.7	107.1	5.8	108.9
			WB	14.5	78.5	3.7	4.8	11.9	82.0	9.7	84.8	7.6	87.5	5.0	91.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.5	46.4	49.8	58.4	65.0	72.3	80.0	82.9	83.0	76.6	66.6	56.3	47.8	(6)
(7)		DBStd	15.19	9.86	10.44	9.30	7.55	6.51	4.38	3.67	4.46	5.66	8.29	9.28	9.47	(7)
(8)		HDD50	554	186	123	33	5	0	0	0	0	0	5	45	157	(8)
(9)		HDD65	2264	580	434	242	89	19	0	0	0	1	81	283	535	(9)
(10)		CDD50	6211	74	116	292	454	693	901	1019	1023	797	519	233	90	(10)
(11)		CDD65	2447	2	7	36	89	246	451	554	558	348	130	22	4	(11)
(12)		CDH74	26906	39	100	343	817	2162	4999	6722	6788	3477	1226	187	46	(12)
(13)		CDH80	13087	4	22	74	235	795	2474	3682	3757	1598	406	35	5	(13)
(14)	Wind	WSAvg	7.0	7.4	8.0	8.7	8.9	7.9	7.0	5.7	5.1	5.1	5.9	6.8	7.1	(14)
(15)	Precipitation	PrecAvg	34.5	1.9	2.4	2.9	3.1	3.9	4.1	2.1	3.0	3.0	3.2	2.4	2.2	(15)
(16)		PrecMax	51.4	3.7	9.2	7.6	7.6	8.7	12.0	4.9	9.2	6.9	8.5	8.1	9.7	(16)
(17)		PrecMin	20.6	0.5	0.0	0.1	0.2	1.9	1.2	0.0	0.2	0.4	0.7	0.3	0.1	(17)
(18)		PrecStd	8.6	0.9	2.1	1.8	2.0	1.7	2.8	1.5	2.2	1.7	2.0	2.0	2.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.2	82.4	86.2	90.9	94.7	99.9	102.5	102.5	99.0	91.6	84.4	79.3	(19)
(20)			MCWB	56.4	59.5	62.2	65.4	70.1	70.4	69.8	70.4	68.7	68.6	64.8	60.3	(20)
(21)		2%	DB	72.7	76.8	81.8	86.0	90.6	97.1	99.4	99.7	94.7	88.2	79.3	73.1	(21)
(22)			MCWB	55.0	59.1	62.0	64.5	70.0	70.5	70.4	70.7	70.0	68.8	62.1	56.8	(22)
(23)		5%	DB	68.0	72.2	78.1	81.9	88.0	94.6	97.1	98.1	91.4	84.4	75.0	69.2	(23)
(24)			MCWB	54.7	57.0	60.7	64.3	69.3	71.0	71.1	71.2	69.9	66.6	60.8	55.9	(24)
(25)		10%	DB	63.3	67.3	73.2	78.9	84.5	91.1	94.8	95.4	89.6	81.3	71.7	63.9	(25)
(26)			MCWB	52.4	55.8	59.1	63.2	68.9	71.3	71.4	71.5	70.0	64.3	59.6	53.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.3	66.1	68.2	71.4	75.4	77.3	76.9	76.3	75.4	73.5	69.1	65.9	(27)
(28)			MCD B	68.2	74.0	77.9	79.9	86.7	89.2	89.9	88.5	86.6	83.3	77.6	72.0	(28)
(29)		2%	WB	60.5	63.5	66.3	69.4	73.6	75.4	75.4	75.3	74.3	71.8	66.3	62.8	(29)
(30)			MCD B	66.8	70.4	75.0	78.8	84.0	87.0	87.7	88.2	85.3	81.5	73.1	68.6	(30)
(31)		5%	WB	57.3	60.8	64.6	68.1	72.4	74.3	74.5	74.5	73.3	70.5	64.4	59.2	(31)
(32)			MCD B	64.1	67.8	72.5	77.2	82.4	85.9	87.0	87.7	84.1	79.1	70.7	64.5	(32)
(33)		10%	WB	53.1	57.4	63.0	66.5	71.2	73.3	73.7	73.5	72.4	68.8	62.6	55.2	(33)
(34)			MCD B	61.9	65.2	70.1	74.9	80.8	84.9	86.3	87.1	82.7	76.8	69.0	63.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	22.3	22.0	22.6	22.3	20.1	20.5	21.9	21.8	21.2	22.9	22.4	22.0	(35)
(36)			MCDBR	29.2	28.8	27.0	26.6	24.2	24.6	25.3	25.2	24.4	25.5	26.3	29.5	(36)
(37)			MCWBR	16.0	15.4	12.1	11.9	8.8	5.9	5.8	5.8	6.8	10.2	13.4	16.6	(37)
(38)		5% WB	MCD BR	22.9	24.3	20.4	20.4	19.1	19.4	20.4	21.1	19.5	20.3	20.3	22.3	(38)
(39)			MCWBR	16.1	16.2	11.2	11.3	8.7	7.2	6.2	5.9	6.5	9.9	13.9	16.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.302	0.309	0.339	0.375	0.409	0.414	0.423	0.414	0.388	0.351	0.324	0.304	(40)	
(41)		taud	2.524	2.512	2.443	2.364	2.331	2.352	2.372	2.376	2.394	2.485	2.492	2.523	(41)	
(42)		Ebn at Noon	294	302	299	290	279	277	273	275	279	283	282	287	(42)	
(43)		Edh at Noon	27	30	35	39	41	40	39	38	36	30	28	26	(43)	
(44)	All-Sky Solar Radiation	RadAvg	966	1150	1499	1809	1918	2154	2174	1981	1666	1320	1017	852	(44)	
(45)		RadStd	101	176	116	153	188	184	147	132	136	199	131	111	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page